

GRASSY LAKE, AB, Canada

WMO: 717880

Lat: 49.8736N

Lon: 111.7328W

Elev: 2569

StdP: 13.38

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.1	-12.2	-23.9	1.6	-17.7	-17.7	2.3	-12.0	31.9	32.7	29.3	26.1	8.2	140	0.699

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.1	90.1	63.6	86.6	62.6	83.1	61.5	66.9	82.8	65.0	81.0	63.4	79.0	8.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.5	89.6	73.4	59.4	83.1	71.0	57.5	77.7	68.8	32.9	82.4	31.4	80.5	30.1	79.2	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.5	25.6	22.4	-26.4	96.1	6.2	2.9	-30.8	98.2	-34.4	99.9	-37.9	101.5	-42.3	103.6		
(5)			-26.6	71.4	6.1	2.3	-31.0	73.0	-34.6	74.4	-38.0	75.6	-42.4	77.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.8	21.1	19.2	32.3	42.5	52.8	60.7	67.0	65.2	56.5	44.1	31.3	19.4		
	DBStd	20.30	15.04	15.82	13.65	9.39	7.62	5.96	5.36	5.90	8.59	9.39	12.94	14.40		
	HDD50	4399	895	864	552	252	58	3	0	0	34	226	565	950		
	HDD65	8331	1360	1283	1015	675	382	155	39	72	275	649	1011	1415		
	CDD50	1773	0	0	3	27	145	324	528	470	230	42	4	0		
	CDD65	231	0	0	0	0	5	27	102	77	20	0	0	0		
	CDH74	4054	0	0	0	22	190	481	1596	1279	467	19	0	0		
	CDH80	1446	0	0	0	4	39	129	628	484	161	1	0	0		
(14) Wind	WSAvg	10.6	12.2	10.6	11.7	11.9	10.8	10.0	8.5	8.4	9.5	11.1	11.5	11.3		
(15) Precipitation	PrecAvg	14.3	0.7	0.5	0.7	1.2	1.9	3.3	1.4	1.3	1.2	0.8	0.6	0.6		
	PrecMax	20.9	1.7	1.1	2.0	2.7	4.0	8.1	4.9	3.1	2.7	2.8	1.3	1.3		
	PrecMin	7.6	0.2	0.1	0.0	0.3	0.3	0.6	0.1	0.1	0.0	0.2	0.1	0.0		
	PrecStd	3.0	0.4	0.3	0.5	0.6	0.9	1.6	1.1	0.8	0.8	0.6	0.4	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.9	54.1	65.0	77.1	84.8	89.7	94.9	92.7	90.6	76.9	64.8	51.2		
		MCWB	41.9	43.2	47.8	52.6	57.5	65.6	65.9	63.7	61.0	55.9	49.6	40.2		
	2%	DB	47.0	47.2	60.5	70.0	79.5	83.5	90.8	89.2	85.1	69.4	57.0	45.6		
		MCWB	38.8	38.7	45.5	50.6	55.7	61.4	65.5	62.6	59.4	52.1	44.9	38.3		
	5%	DB	43.4	43.0	55.8	64.5	74.5	79.4	86.8	85.8	79.6	64.1	52.4	41.9		
		MCWB	36.9	36.5	42.9	47.1	53.7	60.1	64.5	61.8	57.6	49.5	43.1	36.1		
	10%	DB	40.0	39.6	50.7	59.6	69.7	75.6	83.1	81.7	73.8	59.4	48.0	38.9		
		MCWB	35.3	34.4	40.2	44.8	51.9	58.3	63.2	61.0	55.3	47.0	39.7	33.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.7	44.1	49.3	54.0	58.7	67.0	70.6	67.6	63.3	57.1	50.0	42.0		
		MCDB	50.8	53.9	63.2	71.9	79.7	81.1	86.9	84.3	81.5	71.2	63.5	48.6		
	2%	WB	39.3	39.5	46.8	50.8	56.7	64.3	67.9	65.2	60.8	53.3	46.0	38.7		
		MCDB	46.2	46.2	58.9	68.5	74.9	80.0	83.9	82.4	80.0	67.0	55.6	44.5		
	5%	WB	37.3	36.9	43.7	48.0	55.0	61.8	66.2	63.7	58.6	50.4	43.3	36.3		
		MCDB	42.5	42.7	54.9	62.1	71.0	76.2	82.1	80.7	76.5	62.3	51.7	41.5		
	10%	WB	35.3	34.5	40.7	45.4	53.3	59.7	64.5	62.1	56.4	47.9	40.5	34.0		
		MCDB	39.9	39.1	49.9	58.0	67.3	72.9	79.8	78.3	72.7	58.6	47.3	38.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.2	19.6	21.7	23.5	24.4	24.2	28.1	28.2	26.5	23.1	19.6	17.7		
		MCDBR	18.9	21.8	28.1	33.2	33.2	30.7	34.9	35.9	36.3	29.8	25.0	19.0		
		MCWBR	12.9	15.2	15.6	16.1	13.8	13.0	14.3	13.9	14.8	15.3	14.8	13.4		
	5% WB	MCDBR	18.1	21.1	27.5	30.9	28.7	28.5	30.7	31.4	34.6	28.0	24.3	19.0		
		MCWBR	12.9	15.4	16.0	15.3	12.9	13.2	13.9	13.7	14.5	14.7	15.2	13.9		
(40) Clear-Sky Solar Irradiance	taub		0.265	0.271	0.295	0.344	0.363	0.354	0.355	0.377	0.334	0.292	0.265	0.255		
	taud		2.435	2.475	2.464	2.321	2.316	2.394	2.406	2.314	2.415	2.514	2.494	2.421		
	Ebn at Noon		254	283	293	285	284	286	284	271	274	270	254	241		
	Edh at Noon		19	23	29	37	39	36	35	37	30	23	18	16		
(44) All-Sky Solar Radiation	RadAvg		395	710	1128	1559	1850	2031	2214	1777	1266	787	440	308		
	RadStd		36	59	82	108	136	125	111	112	105	79	32	27		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (65 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page